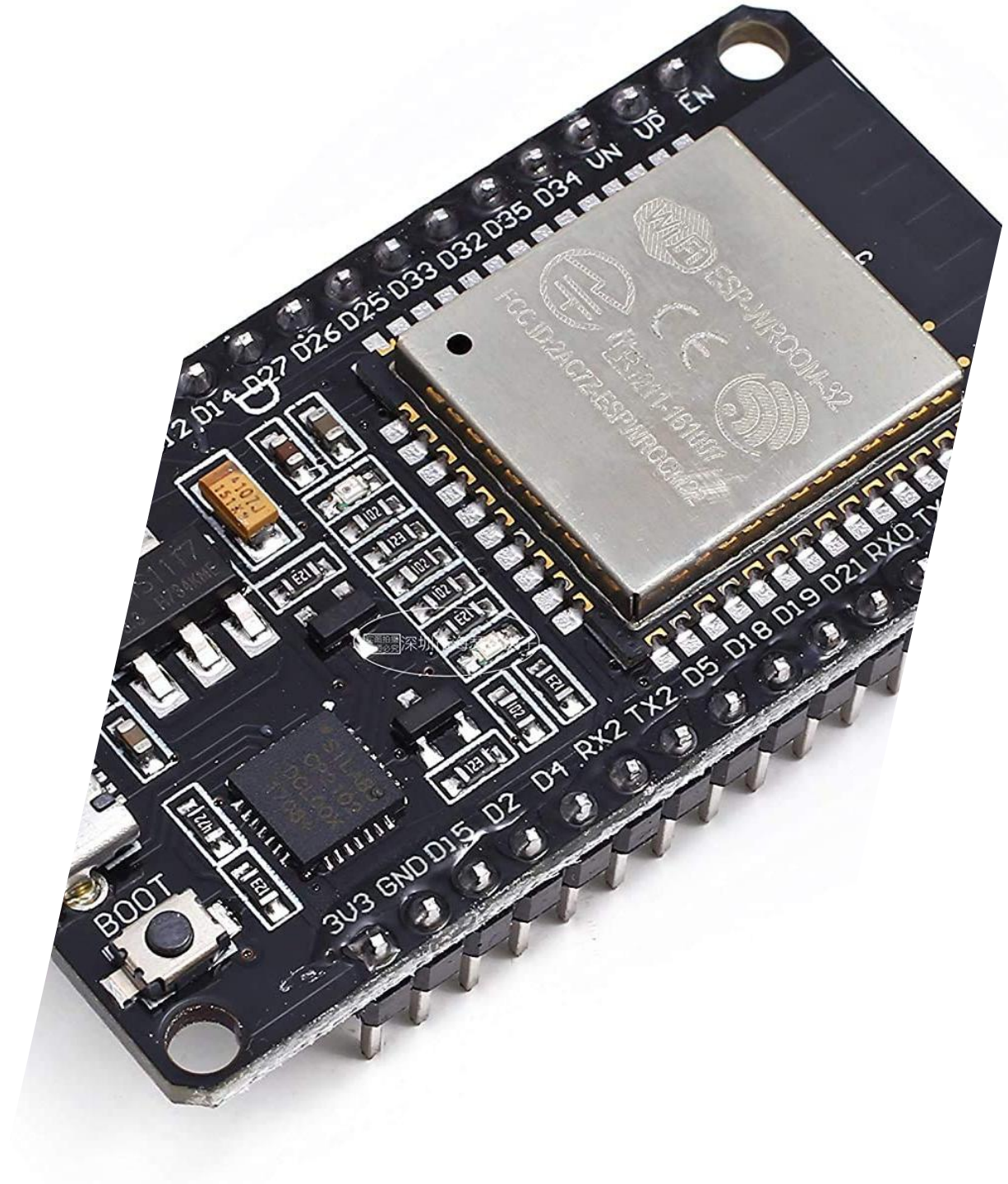


SECURE ESP-NOW

PERSONAL RESEARCH PROJECT

Jovan Christopher



AGENDA

Introduction

Research Questions

Tests and Experiments w/ demo

Implementation w/ demo

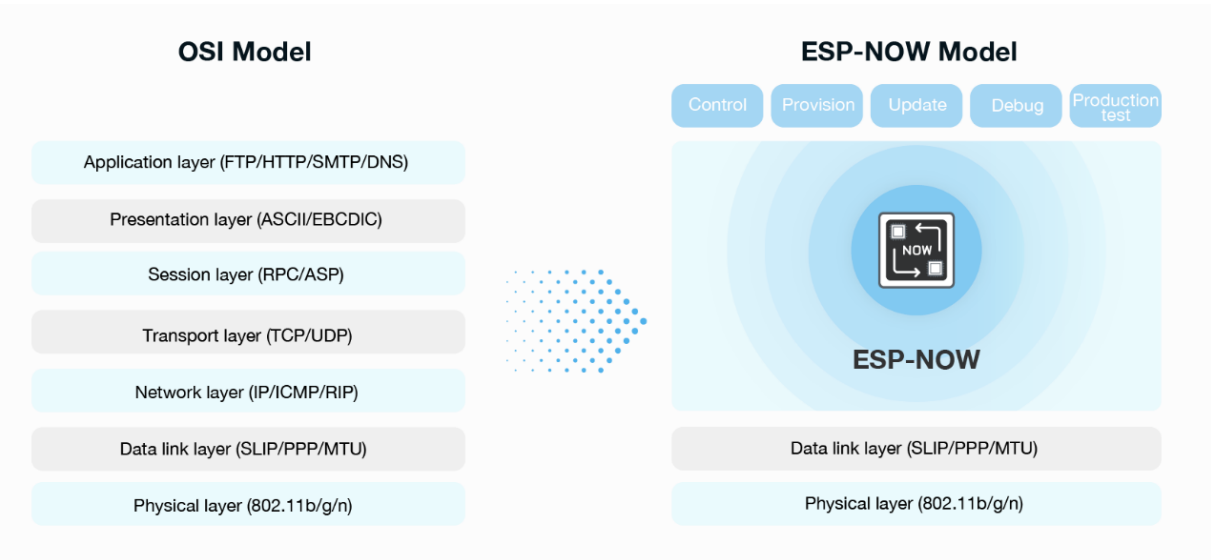
Research Questions Answers

BoK Implementations

Summary and Questions

INTRODUCTION

What is ESP-NOW?



- Wireless communication based on **data-link layer**
- Developed by Espressif Systems
- 2.4 GHz band
- Released in 2022

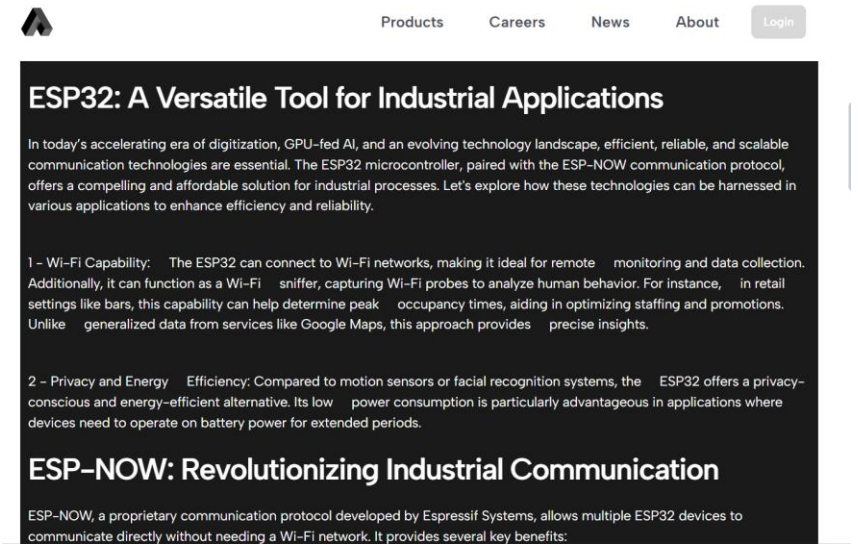


Image source: https://www.aeddix.com/blog/the-rise-of-esp32-and-esp-now-for-industrial-automation

INTRODUCTION

Real-Life Application

- Aeddix Systems GmbH published a news article regarding the use of ESP-NOW



Products Careers News About Login

ESP32: A Versatile Tool for Industrial Applications

In today's accelerating era of digitization, GPU-fed AI, and an evolving technology landscape, efficient, reliable, and scalable communication technologies are essential. The ESP32 microcontroller, paired with the ESP-NOW communication protocol, offers a compelling and affordable solution for industrial processes. Let's explore how these technologies can be harnessed in various applications to enhance efficiency and reliability.

1 – Wi-Fi Capability: The ESP32 can connect to Wi-Fi networks, making it ideal for remote monitoring and data collection. Additionally, it can function as a Wi-Fi sniffer, capturing Wi-Fi probes to analyze human behavior. For instance, in retail settings like bars, this capability can help determine peak occupancy times, aiding in optimizing staffing and promotions. Unlike generalized data from services like Google Maps, this approach provides precise insights.

2 – Privacy and Energy Efficiency: Compared to motion sensors or facial recognition systems, the ESP32 offers a privacy-conscious and energy-efficient alternative. Its low power consumption is particularly advantageous in applications where devices need to operate on battery power for extended periods.

ESP-NOW: Revolutionizing Industrial Communication

ESP-NOW, a proprietary communication protocol developed by Espressif Systems, allows multiple ESP32 devices to communicate directly without needing a Wi-Fi network. It provides several key benefits:

INTRODUCTION

Real-Life Application

BUILD TECH

Revolutionizing Remote Control: Explore Micromote, an ESP-NOW Based Gesture Remote with XIAO ESP32S3 and Smart Gesture Sensor

By Sue Guan 1 year ago

Discover “MicroMote,” a revolutionary device that empowers you to take control of your smart devices with the wave of your hand!

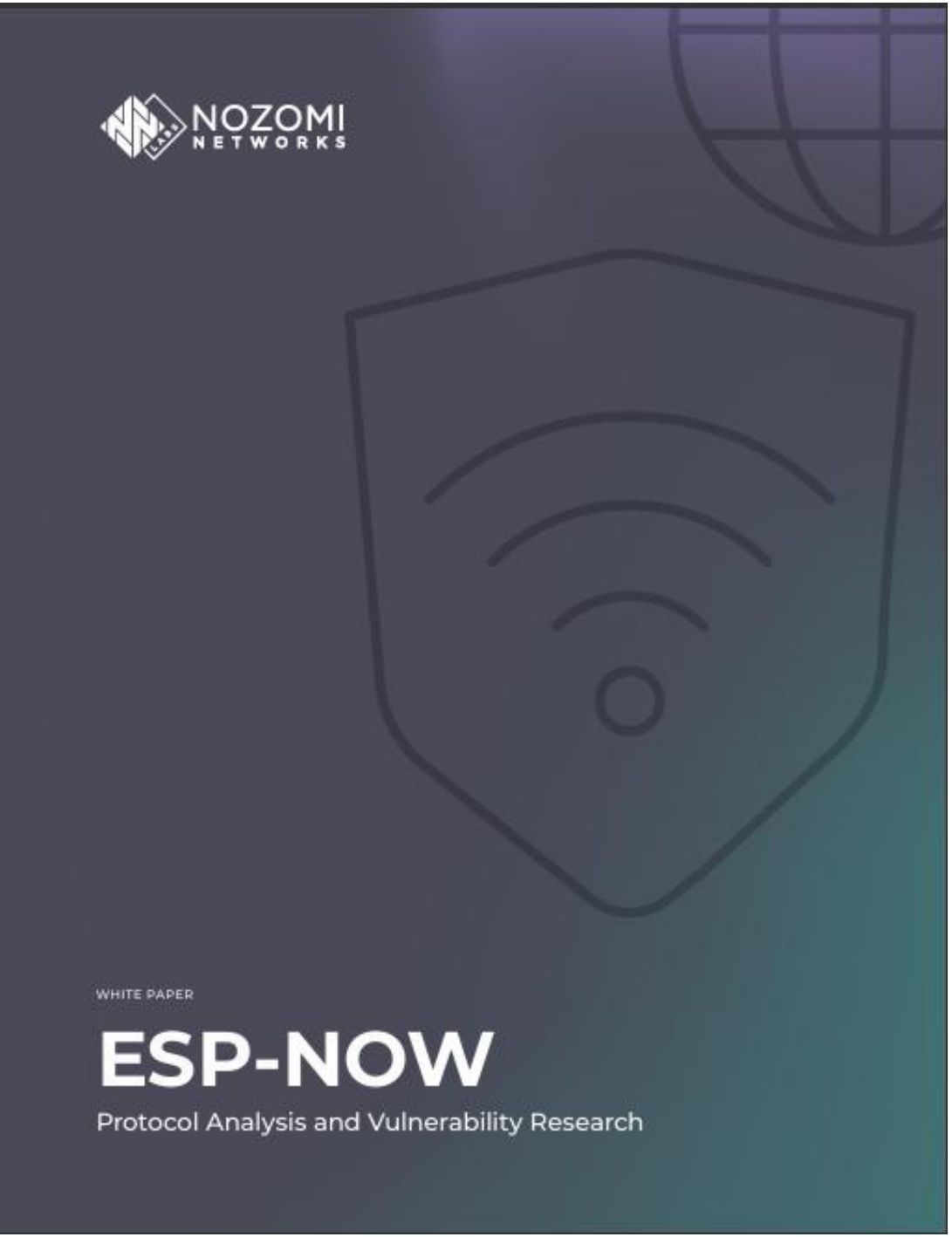
Seeed Hardware: Seeed Studio XIAO ESP32S3, Grove-Smart IR Gesture Sensor

Software: Arduino IDE, Autodesk Fusion 360

Industry: Smart Home, Healthcare, Education, Gaming, Virtual and Augmented Reality

- Seeed Studio created “MicroMote” based on ESP-NOW
- Gesture remote for smart devices

Source: https://www.seeedstudio.com/blog/2023/12/11/revolutionizing-remote-control-explore-micromote-an-esp-now-based-gesture-remote-with-xiao-esp32s3-and-smart-gesture-sensor/



INTRODUCTION

Real-Life Application

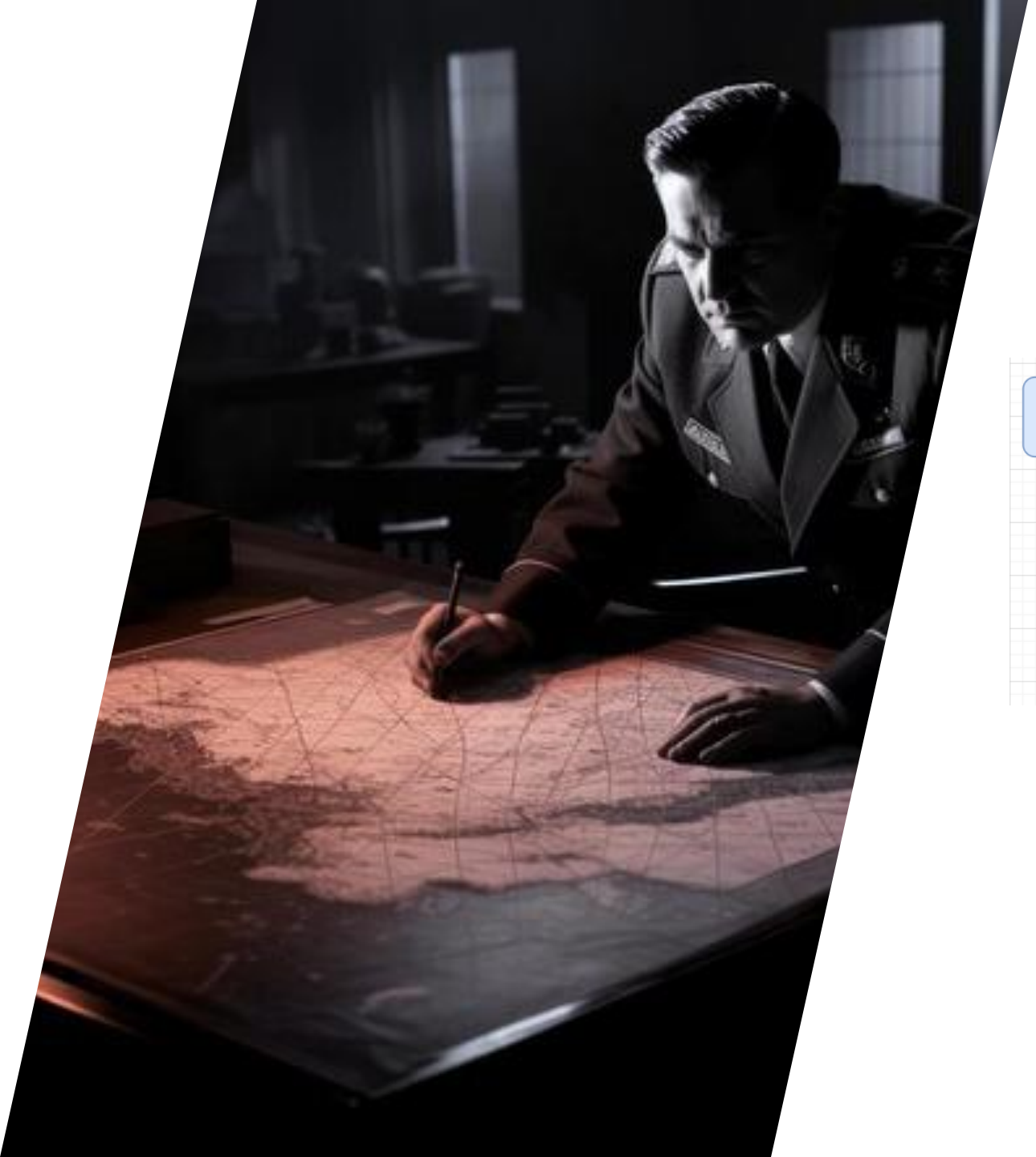
- Nozomi Networks, Cybersecurity company founded in Switzerland
- Published a protocol analysis and vulnerability research on ESP-NOW in September 2024
- Important vulnerability: replay attacks even on encrypted packets (CVE-2024-42483)



RESEARCH QUESTIONS

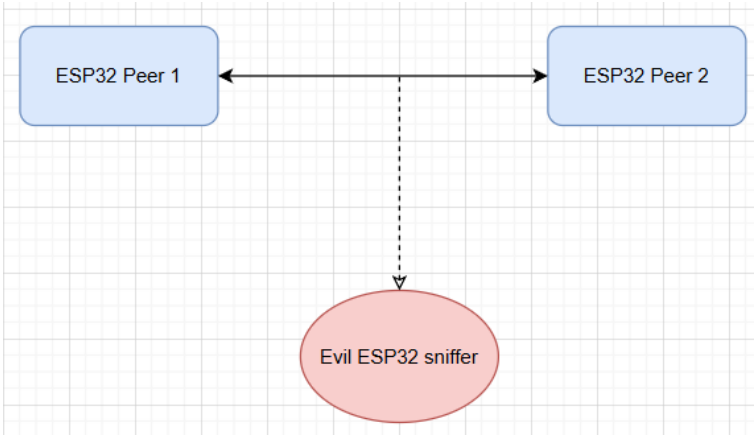
How to improve ESP-NOW reliability by providing a secure transfer of data?(Brainstorm)

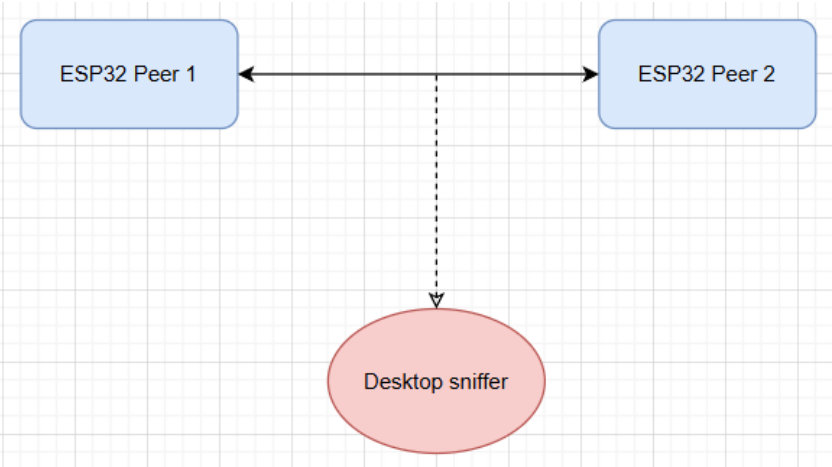
- What are the potential vulnerabilities of ESP-NOW and how to exploit it? (Community Research and Literature Study)
- How to sniff or intercept data being transmitted through ESP-NOW? (Community Research and then implement Security Tests)
- What security measures could be implemented?(Community Research, Literature Study, and Available Product Analysis. Implementation will then be conducted while using System Tests, Code Review, and Prototyping)



TESTS AND EXPERIMENTS

Initial Attack Plans(before testing)





TESTS AND EXPERIMENTS

FIRST PHASE RESULTS

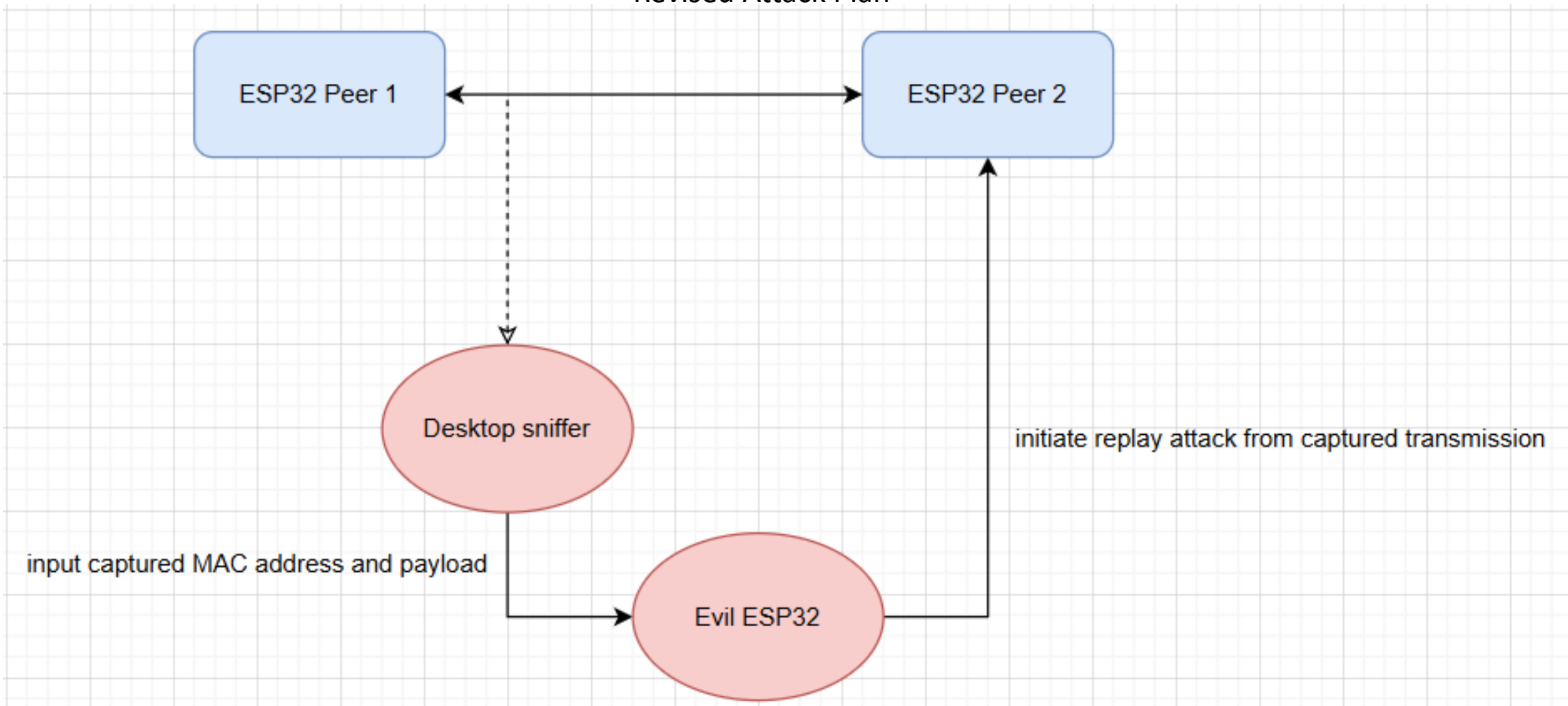
```
Packet received: Length 59
Payload: ..........@".;.8.......7...4rJ......4..Hack Me.........xV.
Packet received: Length 59
Payload: ..........@".;.8.......7...4.0.g....4..Hack Me.........xV.
Packet received: Length 59
Payload: ..........@".;.8.......7...4.<.h....4..Hack Me.........xV.
Packet received: Length 59
Payload: ..........@".;.8.......8...4)*......4..Hack Me.........xV.
---- Closed the serial port COM6 ----
```

```
.000 0001 0011 1010 = Duration: 314 microseconds
▸ Receiver address: Espressif_3b:ed:38 (40:22:d8:3b:ed:38)
▸ Destination address: Espressif_3b:ed:38 (40:22:d8:3b:ed:38)
▸ Transmitter address: Espressif_10:0e:44 (a0:dd:6c:10:0e:44)
▸ Source address: Espressif_10:0e:44 (a0:dd:6c:10:0e:44)
▸ BSS Id: Broadcast (ff:ff:ff:ff:ff:ff)
  .... .... .... 0000 = Fragment number: 0
  0000 0110 0100 .... = Sequence number: 100
  Frame check sequence: 0xdb956c9d [unverified]
  [FCS Status: Unverified]
  [WLAN Flags: ........C]
▸ IEEE 802.11 Wireless Management
```

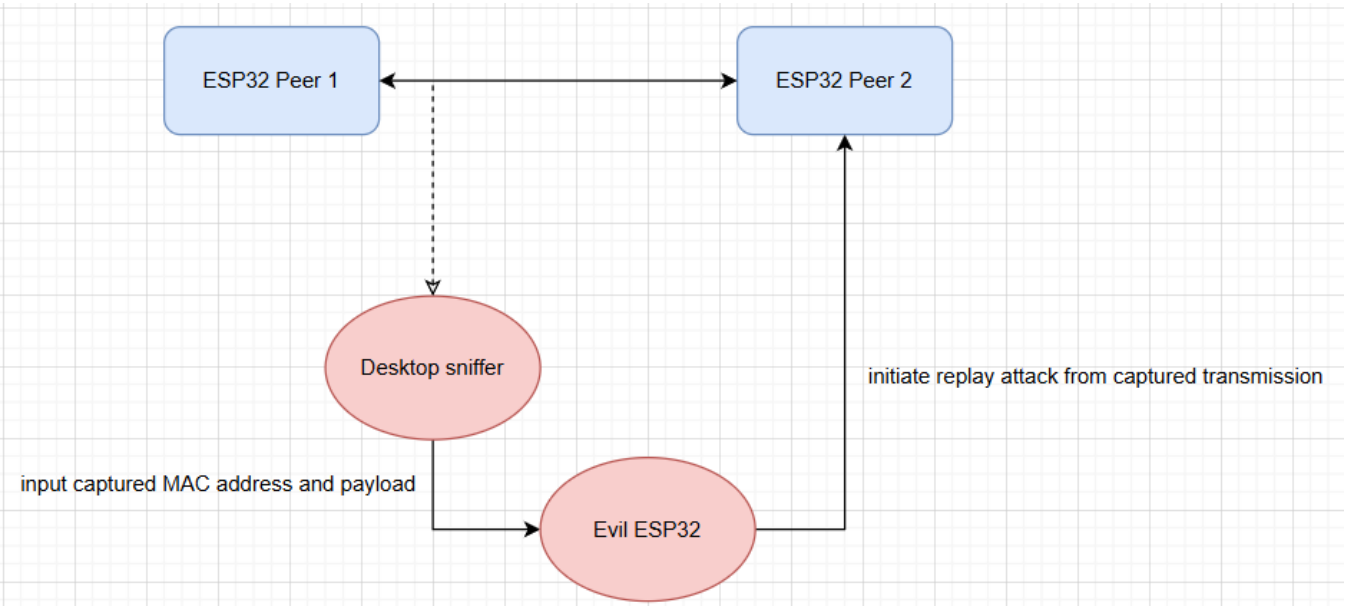
```
0010  00 00 24 00 2f 40 00 a0  20 08 00 00 00 00 00 00   ··$·/@·· ·····
0020  f1 17 a2 7b 00 00 00 00  10 02 71 09 a0 00 c1 00   ···{···· ··q·····
0030  00 00 c1 00 d0 00 3a 01  40 22 d8 3b ed 38 a0 dd   ········:· @".;·8··
0040  6c 10 0e 44 ff ff ff ff  ff ff 40 06 7f 18 fe 34   l··D···· ··@····4
0050  d6 1f d0 34 dd 15 18 fe  34 04 01 48 61 63 6b 20   ···4···· 4··Hack
0060  4d 65 2e 00 00 00 00 00  00 00 88 9d 6c 95 db      Me.····· ····l··
```

TESTS AND EXPERIMENTS

Revised Attack Plan



TEST AND EXPERIMENTS



- Successful replay attack on unencrypted ESP-NOW transmission
- Combination of Wireshark and ESP32

```
---- Opened the serial port COM5 ----
Hack Me.
16
```

```
E (327) esp_core_dump_flash: No core dump partition found!
Target channel:
---- Sent utf8 encoded message: "1" ----
Payload:
---- Sent utf8 encoded message: "Hack Me." ----
Sent

Last Packet Send Status:       Delivery Success
```

IMPLEMENTATION

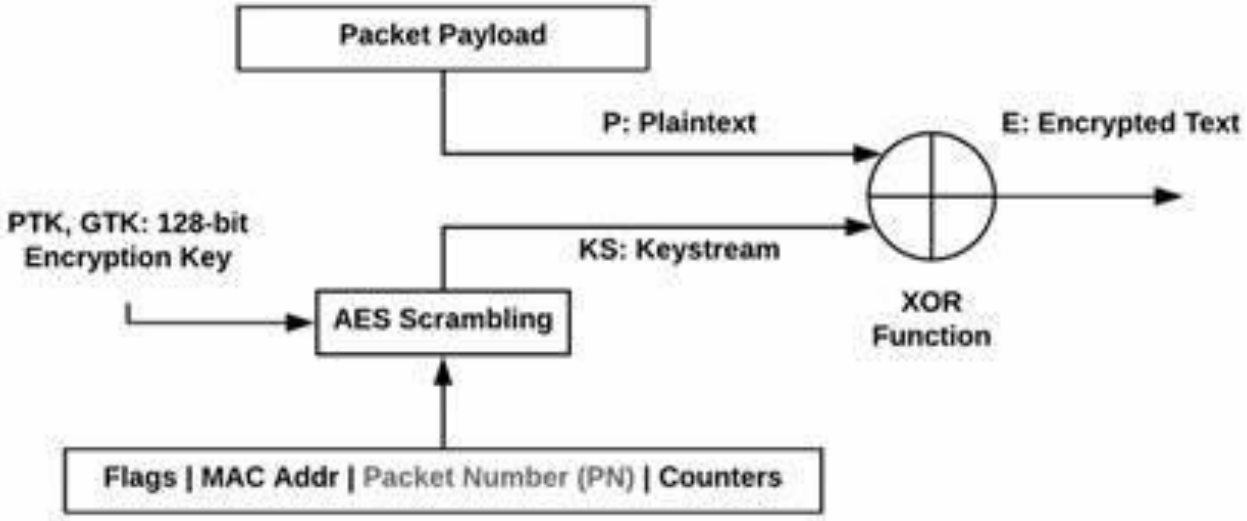


- Using the built-in ESP-NOW encryption(patched)
- According to Nozomi Networks, not long after they discovered the vulnerability, Espressif has patched the vulnerability

Source: https://www.nozominetworks.com/blog/flaws-in-espressif-esp-now-allow-attackers-to-replay-communications

CCMP WITH AES-128

- Standard security provided by Espressif's ESP-NOW library
- Replay attack vulnerability fixed in ESP-NOW 2.5.2
- Providing up to 16 bytes of encryption key(both PMK and up to 6 LMK)





RESEARCH QUESTIONS(ANSWERS)

How to improve ESP-NOW reliability by providing a secure transfer of data?

Implementing message encryption and authentication while making sure that a self-generated encryption key of 16 bytes is used.

- What are the potential vulnerabilities of ESP-NOW and how to exploit it? ***Replay attack vulnerability persists although the risk is lower after the fix, out of bounds memory read. Exploiting encrypted ESP-NOW could be done by intercepting raw packets and retransmitting the packet to the destination node, however this needs deep understanding of the protocol and lots of experience.***
- How to sniff or intercept data being transmitted through ESP-NOW? ***Wireshark or any other network analysis/sniffer software can be used including an ESP-32.***
- What security measures could be implemented? ***Basic security: built-in security by ESP-NOW using CCMP and AES-128. Advanced security: lots of encryption and authentication library to be used in combination with ESP-NOW such as ChaCha20 and GCMP however in the world of embedded systems memory is a scarce resource most of the time so deciding on a protocol that is not overkill is important.***



BOK IMPLEMENTATION

- Network Scanning (Offensive)
- Radio Hacking (Offensive)
- Network Spoofing and MITM (Offensive)
- Common Vulnerabilities and Exposures (Defensive)
- Secure Software Design (Defensive)
- Encryption and Authentication (Defensive)



Summary and Questions

- Importance of encryption and authentication in wireless communication
- Security in IoT
- Implementing a secure data transfer by design